ML4T Development Environment Setup

Revision History

1. Posted at start of term.

ML4T Course Environment

Setting up your development environment correctly is essential for success in ML4T. All students **must ensure their code runs in the course-specific Linux-based environment**, as this is the platform used by Gradescope for grading. Claims like *"it works on my machine"* will not be accepted if your code fails on Gradescope.

We've created detailed **video tutorials** in the **Media Gallery (https://gatech.instructure.com/courses/498126/external_tools/17731) on Canvas** to walk you through the setup and validation process. **You must setup the Development Environment to the specifications found in this post specific to your term. We strongly recommend following the steps in the Development Environment videos (required viewing), including the validation steps demonstrated.**

Platform-Specific Guidance

Mac (Apple Silicon)

Students may use their existing macOS installations - but if they encounter issues we will strongly recommend the use of the UTM Virtual Machine to run a clean instance of macOS. While many students are able to run on their existing macOS installation, w*e will no longer troubleshoot issues on native macOS (see "Other Advisories" below).*

→ Follow the UTM setup shown in the Media Gallery videos.

Mac (Intel)

Students may use their existring macOS installations. However, since support for this platform has been dropped by Apple, we recommend moving to an Apple Silicon or Windows solution.

Windows

Use an **external SSD (strongly recommended for better performance)** or a **Linux VM** (e.g., VirtualBox, Hyper-V, VMWare). Students may also use container solutions (e.g., Docker), but they are not officially supported by the course or instructional staff.

→ Follow the Linux setup or External SSD videos shown in the Media Gallery.

Summary of Requirements

✅ Development Environment

- **Hardware:** Computer with at least 4GB RAM and a CPU clock speed of at least 2.5GHz. The computer should have 50GB (or more) of available storage
- **OS**: Linux (Ubuntu/Debian based), or macOS with Linux in a VM
- **Administrative Rights**: You must have administrative rights on your local machine
- **Python**: 3.10 only
- **Key Libraries**: NumPy, SciPy, Pandas, Seaborn, Matplotlib, PyTest, etc. (Only the specific packages and versions provided in `environment.yml`)
- **Internet:** Reliable high-speed internet connection (e.g., 2+ Mbps internet speed recommended). Must not be a mobile phone internet connection.
- **Environment Setup**: Use Conda; setup instructions are in the videos and included below

❌ Not Supported

- No Native **Windows development** (some components will not run)
- No **Linux for exams** (Honorlock does not support it)

✅ Exams

- Must be taken on **Windows or macOS** (not Linux)
- **Chrome browser** is required for exams

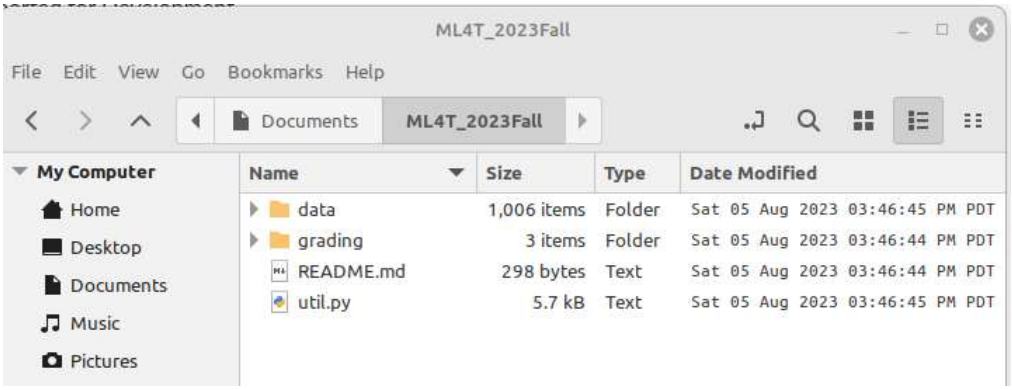
Environment Setup (See setup videos in the Media Gallery)

1. Download ML4T_2026Spring.zip (https://gatech.instructure.com/courses/498126/files/69860837?wrap=1) (https://gatech.instructure.com/courses/498126/files/69860837/download?download_frd=1) (Dropbox (https://www.dropbox.com/scl/fi/0e6exrcztglgcekkvinyh/ML4T_2026Spring.zip?rlkey=w8ck0pchxh7zlk7feuntlrtfr&st=fhjvhdud&dl=0)) and extract it.
→ (Found in Files (https://gatech.instructure.com/courses/498126/files) section of Canvas and Dropbox link in assignments: Files -> ML4T Environment Files -> ML4T_2026Spring.zip)

2. Install Miniconda

3. Save the environment.yml (https://gatech.instructure.com/courses/498126/files/69692729?wrap=1) (https://gatech.instructure.com/courses/498126/files/69692729/download?download_frd=1) file (found in the Files section of Canvas: Files -> ML4T Environment Files -> environment.yml).

Once extracted, you will have a directory structure similar to the one shown below. The .zip files for each project will be extracted into this folder alone, as their relative placement is a critical component of the overall ML4T development environment.



The contents of the environment.yml:

```
name: ml4t
dependencies:
- python=3.10
- cycler=0.11
- kiwisolver=1.4.2
- matplotlib=3.5
- numpy=1.22.3
- pandas=1.4.3
- pyparsing=3.0.4
- python-dateutil=2.8
- pytz=2022.1
- scipy=1.7.3
- seaborn=0.11.2
- six=1.16.0
- joblib=1.1.0
- pytest=7.1.2
- future=0.18.2
- pip
- pip:
  - pprofile==2.1.0
  - jsons==1.6.3
  - gradescope-utils
```

4. Create and activate the environment with the following commands.

```
conda env create --file environment.yml
conda activate ml4t
```

5. macOS / VirtualBox fix (to avoid Matplotlib crash) **-** only needs to be performed once:

For macOS:

```
mkdir -p ~/.matplotlib
echo "backend: TkAgg" > ~/.matplotlib/matplotlibrc
```

For Linux in VirutalBox:

```
mkdir -p ~/.config/matplotlib
echo "backend: TkAgg" > ~/.config/matplotlib/matplotlibrc
```

Validating Your Environment (Very Important)

Use the optional `assess_portfolio` project - the **assess_portfolio.zip (https://gatech.instructure.com/courses/498126/files/69860851?wrap=1) (https://gatech.instructure.com/courses/498126/files/69860851/download?download_frd=1)** file is in the Files section on Canvas and **Dropbox**

(https://www.dropbox.com/scl/fi/968gthq4krw2rz3wyffoo/assess_portfolio_2026Spring.zip?rlkey=6t2gxyyj3rcfttdnurweu3ng8&st=mbybrmjy&dl=0) - to verify your environment:

From `ML4T_2026Spring/assess_portfolio/`, run:

```
PYTHONPATH=../:. python grade_analysis.py
```

Here's an example of the contents of comments.txt for the first assignment using the unchanged template:

```
<pre>--- Summary ---
Tests passed: 0 out of 8

--- Details ---
Test #0: failed
Test case description: Wiki example 1
IncorrectOutput: One or more stats were incorrect.
  Inputs:
    start_date: 2010-01-01 00:00:00
    end_date: 2010-12-31 00:00:00
    symbols: ['GOOG', 'AAPL', 'GLD', 'XOM']
    allocs: [0.2, 0.3, 0.4, 0.1]
    start_val: 1000000
  Wrong values:
    cum_ret: 0.25 (expected: 0.255646784534)
    avg_daily_ret: 0.001 (expected: 0.000957366234238)
    sharpe_ratio: 2.1 (expected: 1.51819243641)

Test #1: failed
Test case description: Wiki example 2
```

The comments.txt file will contain a summary of which tests were passed or failed and any error messages. The points.txt file reports the score from the autograder, used by the teaching staff to automate grading submitted code in a batch run, and can be safely ignored by students.

You must complete this validation step to confirm that your setup is working as intended.

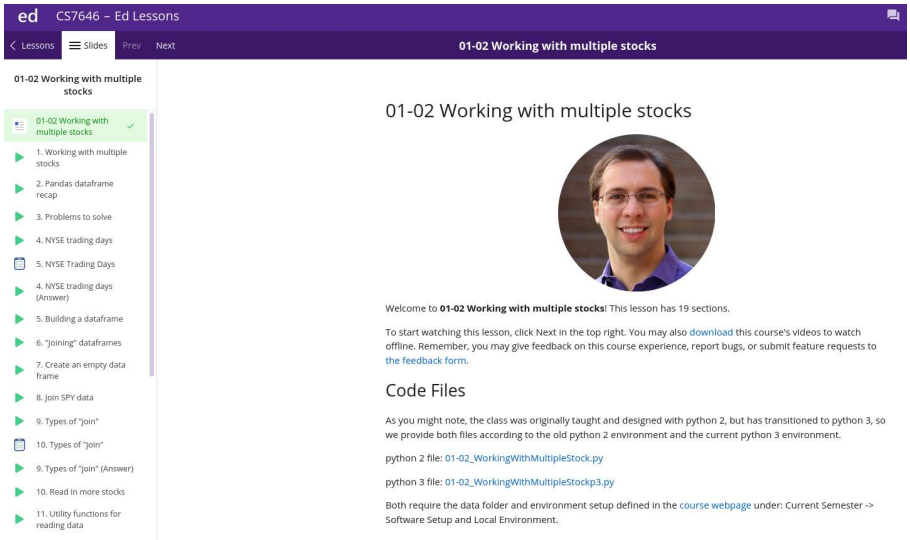
IDE Recommendation

- **Recommended**: PyCharm (JetBrains free student license available at https://www.jetbrains.com/shop/eform/students) (https://www.jetbrains.com/shop/eform/students)
- **Optional**: VSCode
- **Avoid**: Jupyter notebooks and any IDE that installs packages into the `ml4t` environment

Running Lecture Video Code

Early in the term, before students begin to focus on the Assignments, some will want to explore the code associated with the lecture videos. So, this post shares a few thoughts to consider.

1. The .csv files mentioned in the videos are found in the data folder in the ML4T directory once the .zip file is extracted.
2. The use of util.py assumes a specific relative relationship between the location of the .py file relative to the location of the .csv file being accessed. The best way to maintain this relationship is to create a "lessons" folder at the same level as the data and grading folders. The .py files (which can be found in Ed Lessons) should then be placed in the lessons folder without any further nesting of directories. Note that you may need to change the lines of code that read `data/[symbol].csv` to `../data/[symbol].csv` to ensure correct path to the .csv file. This maintains the necessary directory structure and is consistent with the usage found in the course programming assignments.
3. The Ed Lessons videos were recorded when the course used Python 2.X. So, the current objective should be focused on understanding the key concepts, recognizing that some of the specific functions and methods may differ in Python 3.X. To best understand the differences between Python 2 and 3, please consider searching the Internet. As a reminder, excluding the videos, all aspects of our course use Python 3.10.
4. You can download the .py files using Ed Lessons. For example:



Troubleshooting Tips

- **Search Ed and the web** first
- Still stuck? Post on Ed Discussion:
  - Include **video timestamp**
  - Describe the issue
  - Include a **screenshot**

Final Notes

- **Do not modify, move, or copy** `util.py`, `grading/`, or `data/` folders
- **Always** download assignment starter files from Canvas for each project
- Back up your code using **private** GitHub (e.g., github.gatech.edu), Dropbox, or OneDrive
- Students who lose access to their computers are expected to recover (following these instructions to recreate the environment and clone their backup) within 2 days
- Code and reports must not be shared with other students

Other Advisories

macOS with prior Anaconda installations

If one is having issues with the Python version or seeing the Conda ml4t environment in Python even though it is visible in the Terminal:

- Update the Anaconda installation
- **Uninstall (https://docs.anaconda.com/free/anaconda/install/uninstall/)** and Reinstall Anaconda
- **Uninstall (https://docs.anaconda.com/free/anaconda/install/uninstall/)** Anaconda and install miniconda
- Check for login script aliases and remove them (e.g., alias python=python3, alias python=/usr/local/python3, or something similar)
- Create and use a new user in macOS.
- Use a clean macOS instance in a VM. On Intel-mac, consider using VMWare Fusion, Parallels, or VirtualBox. On Apple Silicon, consider UTM (free) or Parallels.

Windows Virtual Machines run slow

- Run the Virtual Machine as an administrator
- Disable Windows Hyper-V
- Use the latest versions of the VM Software (e.g., VirtualBox, VMWare Workstation Pro)
- Try using Microsoft's Hyper-V Manager or use an external Linux SSD

Getting Help

If you encounter problems while creating your environment, be sure to search the forum and Internet for similar errors. If that is unsuccessful in resolving the issue, add a post in the Ed Discussions forum **and state the timestamp in the Setup Video of the step being performed**, a description of the problem, and a screenshot of your OS screen.